Devadath Sabarish

co24btech11006@iith.ac.in | devadathsabarish@gmail.com | +91 6364814117 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.42
Sri Chaitanya Techno School	Class XII (CBSE)	2024	93.20%
New Horizon Gurukul	Class X (CBSE)	2022	89.60%

SCHOLASTIC ACHIEVEMENTS

- **Academic Excellence Award:** Received for having **the highest CGPA** in my branch.
- **Optiver Trade-a-thon:** Got into the **Top 100** amongst the participants of IITH.
- **AlgoUniversity Hackathon 2025:** Qualified for **Stage 2**, ranking in the **top 8% of 250,000+ participants** nationwide.
- **IELTS Examination:** Achieved an overall Band Score of **8.0/9.0 (CEFR Level C1).**

WORK EXPERIENCE

AltoLabs | Engineering Intern

- Integrated multiple **Arabic TTS and STT models** into a unified **FastAPI-based speech platform**.
- Improved **model robustness** by resolving **timeouts, memory issues, and dependency conflicts**.
- Fixed **routing, inference, and backend integration issues** across environments.
- Created **architecture diagrams, status reports, and technical documentation**.

PROJECTS

xv6 Operating System | C

- Extended **xv6** with custom system calls for process, scheduler, virtual memory, disk, and RAID/swap introspection.
- Replaced round-robin with a **syscall-aware MLFQ scheduler** using variable quanta, demotion, and boosting.
- Built **demand-paged virtual memory** with lazy allocation, page faults, **Clock replacement**, and per-process stats.
- Implemented **disk-backed swap** to evict and restore pages under memory pressure.
- Added **disk scheduling** and **RAID 0/1/5-style swap backing** with FCFS/SSTF and failure tests.

Networked Co-op Rowing Game (Glitch Tenure Project) | Unity, C#

- Debugged the **multiplayer rowing gameplay**, including lobby flow, paddle input, disconnect handling, and scene transitions.
- Improved **save/load reliability**, fixing checkpoint restoration, progress mismatches, spawn loading, and quit/disconnect saves.
- Added/fixed gameplay features including **checkpoints, jump pads, final-level timer, and end-game flow**.
- Polished the player experience with **credits, audio integration, fog, shader fixes, custom cursor/icon, and final bug fixes**.

Molecular Dynamics Engine | Python

- Contributed to a team-built **Molecular Dynamics simulator** that models noble-gas melting using Lennard-Jones interactions.
- Optimized force computation with **NumPy vectorization** and **Verlet neighbour lists**, reducing all-pairs distance checks.
- Implemented **cutoff radius**, **skin radius**, and **list rebuild logic** to make simulation steps faster without losing accuracy.
- Built stability checks using **energy conservation**, drift plots, and naive-vs-optimized force/energy comparisons.
- Extended **RDF-based coordination number analysis** to measure multiple neighbour shells and validate FCC

structure.

General ODE Solver | Python

- Contributed to a team-built **ODE boundary value solver** using the **shooting method**.
- Built **accuracy-analysis tools** comparing RK2, RK4, ABM2, and ABM4 across multiple step sizes.
- Generated **Numba-accelerated reference solutions** and used them to validate numerical results.
- Created **error diagnostics and plots** for absolute error, convergence behaviour, and max/RMS error summaries.

Mini-Projects | Verilog

- Completed **Verilog mini-projects** as part of relevant Digital Circuits coursework.
- Designed a **4-bit unsigned ripple-carry adder** using only NAND gates, each modeled with a 1 ns delay.
- Built an **8-bit Fibonacci sequence generator** using D flip-flop registers and ripple-carry adders.
- Implemented **behavioral and structural models** with NAND-based adders, gate-delay analysis, and waveform simulation.

POSITIONS OF RESPONSIBILITY

Game Development Head @ Glitch, The Game Development and E-Sports Club of IIT Hyderabad

Game Development Core @ Glitch, The Game Development and E-Sports Club of IIT Hyderabad

SKILLS

- **Programming Languages and Libraries:** C/C++, Python (NumPy, Matplotlib, Pandas), HTML/CSS*, Verilog*
- **Tools:** GitHub, LaTeX, Linux Shellscript*, Blender, Solid Edge*, Unity

RELEVANT COURSES

- **Programming and Computational Methods Courses -** Introduction to Programming, Data Structures and Applications, Operating Systems, Computer Aided Numerical Methods (A+), Fundamentals of Scientific Computing, Computational Methods in Materials Science, Electronic Structure and Atomistic Modelling, Legacy Responsive Web Design V8*
- **Electronics Courses -** Digital Circuits, Introduction to Hardware Description Language
- **Mathematics Courses -** Introduction to Probability & Statistics (A+), Elementary Linear Algebra, Transform Techniques, Differential Equations (A+)

EXTRACURRICULAR

Participated in the **GameDev Problem Statement** of the **Inter-IIT Tech Meet 14.0**.